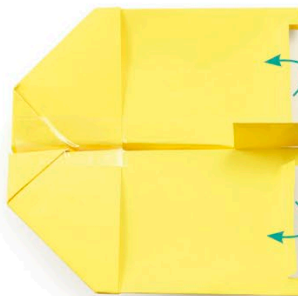




Pinch the tail fin at the top, to ensure a smooth crease.

6 Cut from the center fold up to the base of the wings—about 1 in (2.5cm) in from the back of the plane—then push it up. This is the tail fin.



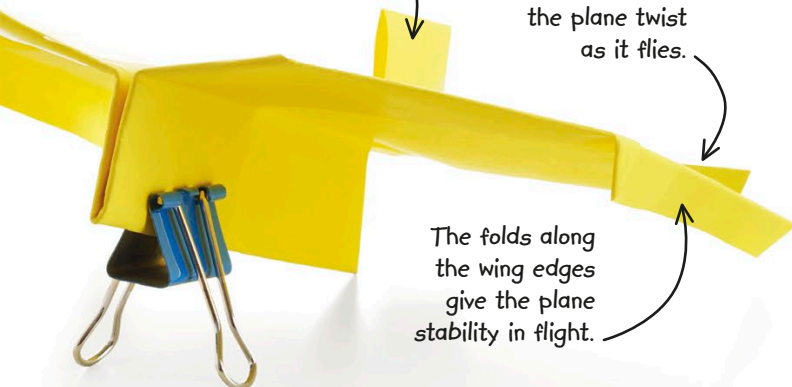
Try to make both slits on either side of the tail fin the same size.

7 Crease the tail fin so it stands up, taping the wings to keep it closed. Cut slits into the trailing edge of each wing and fold up the resulting flaps.

Bending the tail fin to the left or right will make the plane turn in the air.

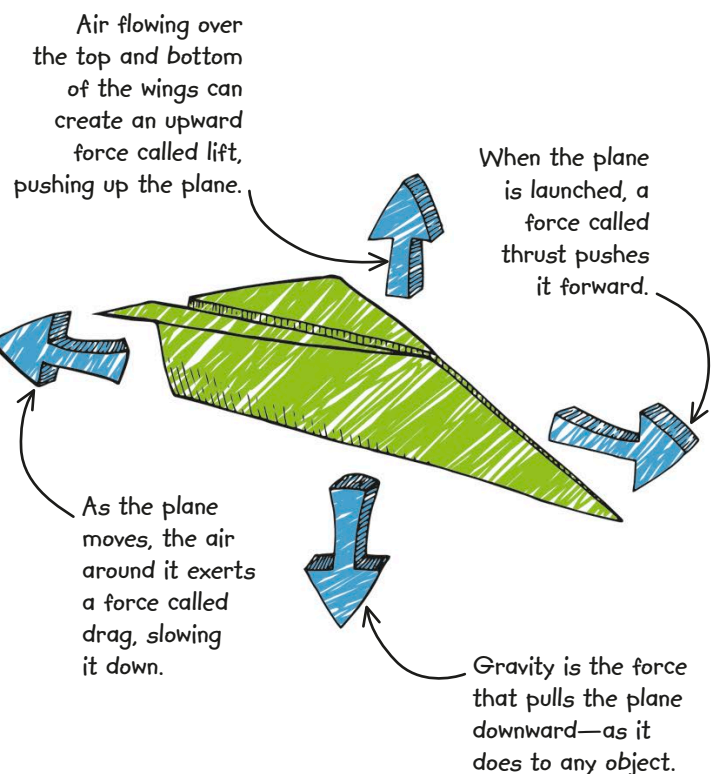
Folding one flap up and the other down will make the plane twist as it flies.

The folds along the wing edges give the plane stability in flight.



HOW IT WORKS

Flying objects experience four types of force: gravity, lift, thrust, and drag. The Dashing Dart speeds through the air because air passes around its streamlined shape easily without exerting much drag. The Graceful Glider experiences lots of lift because its wings have a large area—so it stays in the air for longer. The control surfaces at the back of the Super Stunt Plane change the airflow, creating lift forces that can act sideways or even downward, allowing the plane to change direction, or even spin.



REAL WORLD SCIENCE HANG GLIDER



In the right conditions, with warm air rising up from the ground, a hang glider can stay up for hours. The rising air, called a thermal, pushes up on the underside of the wings, providing lift. To steer, the pilot shifts his or her body to tilt the glider.